

Specified perception interfaces

Driver camera
Emotion / state
network

Road camera
Vehicle detection
and tracking

Vehicle signals
Kinematic estimation

Driver score E
Uncertainty u
Availability a

Hazard score H
Uncertainty u
Availability a

Telemetry score T
Uncertainty u
Availability a

Normalize inputs + apply quality gates

Causal memory + contextual products

Calibrated scalar fusion
UATIF / matched MLP or HGB

Probability + threshold flag + availability status